

Quang Phu Nguyen

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AI Engineer | LLM agents, RAG, agent governance | Vietnamese (native), English (IELTS 6.5)

EDUCATION

Ho Chi Minh City University of Technology (HCMUT)

Ho Chi Minh City

B.Sc. Computer Science (Applied AI) — Graduated, Excellent, GPA 3.8/4.0

Sep 2022 – 2026

- Academic Incentive Scholarship (4/8 semesters), OISP Scholarship (3/8 semesters), Consolidation Prize at the Bach Khoa Innovation Contest.
- Honorable Mention, 2023 ICPC Vietnam Northern Provincial Programming Contest.
- Coursework: AI, data structures and algorithms, computer architecture, mathematical modeling, databases, operating systems.

EXPERIENCE

AI Engineer — FPT Software AI Center, Ho Chi Minh City

Nov 2025 – Present

Agentic ERP Platform · team of 15, stealth mode, ~1,000 users / ~100 DAU

Jun 2026 – Present

- Platform where autonomous agents act inside business systems. Focus: the **agent core** (agents, harness, tools, integrations, prompts, memory, Skills, agent creation), open to UX, data intelligence, trust and governance.
- My scope: agent runtime, permissions and governance, multi-tenant administration, integrations and customer-facing interfaces.
- Built the runtime **default-deny permission layer** so a restricted agent cannot act outside its boundary; delivered per-customer admin controls with a full audit trail and usage/cost reporting for admins and users.
- Wrote the permission-model specification through many review rounds, and kept customer help docs aligned with shipped behavior.
- Shipped the first release of user-authored **Skills** (authoring, validation, create/use interface) and team sharing (view or edit) of agents, skills and artifacts.
- Consolidated user and group management onto one source of truth and migrated legacy permission data onto it.
- Built a third-party integration from scratch and made integrations query remote systems more efficiently.
- Improved chat: conversation search, composer usable while replies stream, ready/failed tab indicator; added queued messaging; made scheduled runs distinguishable from human conversations.
- Operate services with OpenTelemetry, Prometheus, Sentry and Arize Phoenix inside CI quality and architecture gates; review peers on architecture boundaries, permission correctness and database-migration safety.
- Gave three internal AI4ALL sessions (AWS Strands agents, a portal for agent projects, agent-as-a-judge) and mentored an intern.

Stack: Python, FastAPI, SQLAlchemy, PostgreSQL, Redis, Temporal, Qdrant, GKE, ArgoCD, OpenTelemetry, Arize Phoenix, SonarQube, Backstage, React, TypeScript

Healthcare Agentic Chatbot & Management Portal · team of 10, ~100 users

Nov 2025 – May 2026

- Built the core **agent harness from day one** for four domains (product recommendation, health & lifestyle, product education, technical support); optimized execution-plan accuracy, latency and response quality.
- Reached **>90%** accuracy on a golden dataset the tester, BA and PM built from customer discussions.
- Cut typical per-query response latency **30s → 15s** by benchmarking Claude Haiku 3.5, Sonnet 4 and Sonnet 4.5: accuracy across all test cases vs. answer quality (conciseness, helpfulness, instruction following, correctness).
- Cut it to **8s** by replacing agent self-discovery with search-and-rank that narrows domain, use case, tools and context before the agent reasons.
- Solo-built the **Management Portal** from the team lead's idea, beyond contract scope: CRUD and live dashboards (sessions, usage, uptime, latency), knowledge-file approval, prompt management, automated retrieval and scenario tests, audit log.
- The portal helped the customer accept POC-to-production **about two weeks earlier** than on the previous contract; the team won the **“Best Team”** award within FPT Americas.

Stack: Python, FastAPI, Strands Agents, Bedrock (Knowledge Bases, Prompt Management, Guardrails), Lambda, ECS/ECR, CodeBuild, OpenSearch Serverless, React, TypeScript

Smart Product Recommendation System · team of 10

Oct – Nov 2025

- Joined this project on arrival, before the team moved on to the agentic chatbot.

- Rule-based and LLM hybrid recommendation engine for a skin-health scanning device, combining customer data, scan scores, health goals, product catalog and market rules.
- Added latency and quality monitoring, regulatory-compliance checks and automated quality gates for production release.

AI Engineer Intern — FPT Software AI Center

Jun 2025 – Oct 2025

- **M3ARAG** (team of 2): locally deployable, GPU-accelerated multi-agent RAG over PDFs, HTML, Office files and text (LangGraph, Docling, ColPali, quantized local inference), no cloud dependency; open source on GitHub.
- Blog system (team of 2) with hybrid retrieval (BM25 + vector + semantic + freshness scoring) on Azure AI Search and Cosmos DB, Redis caching and fuzzy author search.
- Dual Attention Model (team of 2) extracting technical keywords from company websites, plus a Company–Patent alignment pipeline from scraping to evaluation (with The University of Tokyo).

Research Assistant — HCMUT

Jun 2024 – Dec 2024

- Designed an unsupervised framework to build a Knowledge Graph with minimal domain-expert input.
- Evaluated LLMs for entity and intent extraction from raw Vietnamese documents; studied embedding, dimensionality-reduction and clustering methods to reduce bias in abstract entities.

HONORS & CERTIFICATIONS

Top 10, AI Riser Vietnam 2026 (Google): Sách Cửa Em, 4th of 10 in audience vote	Oct 2026
AI1 Young Talent , Level 3 (FPT Software)	Sep 2026
“Best Team” award, FPT Americas (ST25)	2025
Anthropic: Model Context Protocol, Advanced Topics; AI Fluency; Claude with Vertex AI	May 2026
Google: AI Specialization; Gemini Certified University Student	Dec 2025 – Feb 2026
DeepLearning.AI: AI Agents in LangGraph; Build AI Apps with MCP Server	
DataCamp: AI Engineer for Data Scientists, Associate	Sep 2025

PERSONAL PROJECTS

Tech News Mystery <i>LangGraph, Bedrock AgentCore, Qdrant, Terraform</i> Link	May – Jun 2026
• AI tech-news workspace: crawls articles (Crawl4AI, NewsAPI, Tavily), embeds them into Qdrant and groups them into topic clusters on a PCA embedding map. • Answers through a streamed chat agent and a voice agent (ElevenLabs STT/TTS over LiveKit, talk-over interruption); agent runtime on LangGraph and AWS Bedrock AgentCore. • Background work in Celery; infrastructure as Terraform on AWS with GitHub Actions CI/CD; Next.js front end.	
Fine-tuning LMs for NLP <i>T5, BART, LoRA</i> Link	Apr 2025
• Compared T5-Base, BART-Base and Flan-T5-Small on sentiment, QA and translation (SQuAD, IMDb, WMT En–De): full fine-tuning vs. LoRA vs. adapters, with FP16 and gradient accumulation (5-member team).	
Sentiment Analysis with Various Models <i>scikit-learn, XGBoost, Bi-LSTM</i> Link	Apr 2025
• Benchmarked Decision Trees, Naive Bayes, SVM, XGBoost, Random Forest, MLP and Bi-LSTM over TF-IDF, bag-of-words and embedding features (5-member team).	
Detect AI-generated Text <i>DistilBERT, scikit-learn</i> Link	Dec 2024
• Compared SVM, Random Forest and XGBoost with DistilBERT to classify LLM-written essays in the educational domain (3-member team).	

TECHNICAL SKILLS

Agents & LLM: LangGraph, LangChain, AWS Strands, MCP, RAG, tool calling, agent evaluation, Docling, ColPali, OpenAI and Gemini APIs

ML: Python (4 yrs), PyTorch, TensorFlow/Keras, Hugging Face (PEFT/LoRA), scikit-learn, XGBoost, Sentence-Transformers, spaCy

Backend & data: FastAPI, Pydantic, SQLAlchemy/Alembic, PostgreSQL, Redis, Temporal, MongoDB, Qdrant/ChromaDB, React/TypeScript

Cloud & ops: AWS (Bedrock AgentCore, Lambda, ECS), GCP (GKE), Azure (AI Search, Cosmos DB), Docker, Kubernetes, ArgoCD, Terraform

Quality & observability: OpenTelemetry, Prometheus, Sentry, Arize Phoenix, SonarQube, Playwright; default-deny permissions, forward-only migrations, staged releases